

The Stephensons

Although the first steam locomotive was built in 1804, it was the British father-and-son engineers George and Robert Stephenson who made steam railways a reality. They built the first public railway in 1825 (see p.126) and created the famous *Rocket* (see p.127)—a locomotive that pulled trains on the first passenger railway, between Liverpool and Manchester in the UK, when it opened in 1830.

► FATHER AND SON AT WORK

The Stephensons were practical engineers, who had the foresight and inventiveness to realize the potential of railways.



SAFETY LAMP

Candles can easily set off explosions in mines because of flammable gases in the air. So in 1818, George Stephenson and the British scientist Humphry Davy both came up with designs for safety lamps for miners. In these designs, the flame would be kept covered, out of reach of the gases in the mines. Stephenson's lamp was soon widely used in mines across the northeast of England until the advent of electric light.

Safety lamp based on George Stephenson's design

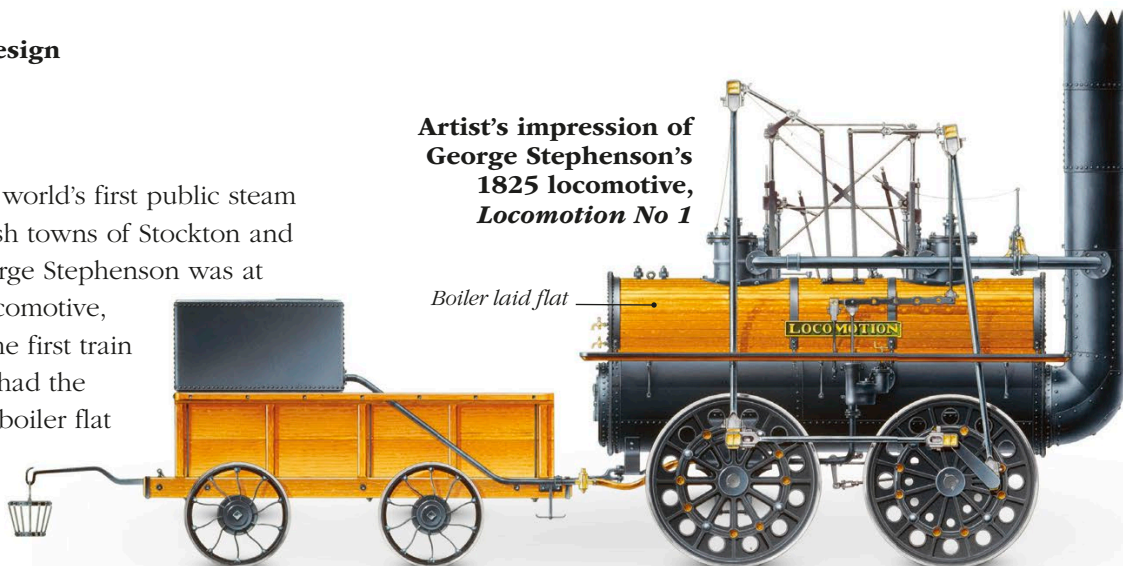
ENGINEERS AND DESIGNERS

As well as designing tracks and engines, the Stephensons were also bridge engineers. Robert Stephenson's design for the railway bridge across the Menai Straits in North Wales, UK, was revolutionary. The trains ran inside two box-shaped iron tubes, supported by huge brick pillars. With the load of the trains spread through all four sides of the boxes, it was very strong. Stephenson's design was hugely influential and is still used in some bridges today.

LOCOMOTION

The Stephensons built the world's first public steam railway between the English towns of Stockton and Darlington in the UK. George Stephenson was at the controls when their locomotive, *Locomotion No 1*, pulled the first train in 1825. The Stephensons had the brilliant idea of laying the boiler flat to deliver power to the wheels more effectively.

Artist's impression of George Stephenson's 1825 locomotive, *Locomotion No 1*



LIFE STORY

1781

George Stephenson is born in Northumberland, UK. His parents are too poor to send him to school, so he starts working at a local mine.

1803

George's son, Robert, is born. Robert's mother and baby sister die the following year.

1818

George Stephenson invents a safety lamp for miners, which provides light while avoiding the danger of flammable gases in mines catching fire.

1823

The Stephensons make a company to develop steam railways and set up the world's first locomotive works in Newcastle, UK.